Al lie between the respective data from alloys without and with 8 wt.% Al. The alloys with Al additions of 4 and 6 wt.% especially are only very weakly affected by the applied ageing treatments in terms of their mechanical data compared with the alloys Fe– 30Mn–1.2C (weakening /embrittlement) and Fe–30Mn–1.2C–8Al (strengthening). […]

4.2 Microstructure of the RAP samples

[…]. Microstructures in the as-homogenised state (unaged) are shown on the left-hand side of Fig. 5, those after ageing at 550°C for 24 h on the right-hand side, respectively. In the as-homogenized state the three alloys exhibit almost identical austenitic microstructures with an average grain size of about 80 lm, few twins (increasing with higher Al content) and no apparent microsegregations. After ageing, however, pronounced differences between the three alloys could be observed. Coarse particles with a diameter of ~10 μm appear at the grain boundaries of the Al-free alloy (Fe–30Mn–1.2C), most probably consisting of a pearlitic ferrite/(Fe,Mn)3C microstructure. The addition of Al to the alloy Fe–30Mn–1.2C–4Al apparently constrained the formation of those phases during ageing, as the number density and size of the particles was now significantly lower than in the Al-free material and only thin films appeared on the grain boundaries of the alloy with 4 wt.% Al. A further increase in Al content (alloy Fe–30Mn–1.2C–8Al) resulted in the complete absence of grain boundary particles during ageing, but a large number

of unevenly distributed small particles appeared within the grains, giving them a darker contrast.

4.3 Tensile testing of the conventionally produced samples

For reference and comparison Fig. 6 shows exemplar engineering stress–strain curves for the alloys Fe–30Mn–1.2C–2Al (red) and Fe–30Mn–1.2C–8Al (blue) after ageing at 450°C for 1 h, obtained by following the conventional synthesis and processing (Fig. 6a) and RAP (Fig. 6b) approaches. For the alloy Fe–30Mn– 1.2C–8Al both the RAP and conventional data are in the same range, with an only lightly increased YS (610 to 540 MPa), lower UTS (810 to 890 MPa) and almost identical ductility (76 to 73%) of the RAP samples. While this trend (slightly higher YS and lower UTS) is the same for the alloy Fe–30Mn–1.2–2Al, the RAP samples now exhibited significantly less ductility than the conventional specimens (49 compared with 84%) under this specific material condition. The unusually large elastic strain that is apparent in Fig. 6b can be attributed to slight deformation (bending) of the RAP samples, which could not be completely avoided despite the clamping procedures applied during quenching of the thin segments.

3.4 有用的词汇和短语

3.4.1 语言任务

练习3：阅读本单元的结果部分和你的目标研究论文。给可以使用在以下四个方面的词汇或短语画线或做标记，并将你的建议与3.4.2节中的列表进行比较。

1. 对结果的主观描述，例如 extremely、often

2. 将结果与模拟/预测结果/其他研究结果进行比较的语言，例如 in line with

3. 处理结果问题的语言，例如 unfortunately

4. 表示结果含义的语言，例如 suggesting that

3.4.2 结果部分使用的语言

本节列出通过分析2500多篇不同学科已发表的研究论文所得到的用于结果的词汇和短语。这个列表仅包含经常出现的词汇和短语，因此被作者和编辑认为是常规且可以接受的。

该列表还会有助于你保持写作的流畅性。在每个列表的下面都有一些示例以说明词汇和短语在句子中的使用方式。因此，当你感到困惑并且想不出该写什么的时候，请查看这里的列表和句子示例。

观看结果的邀请

according to the data in Fig. 1,	**Verbs**
as can be seen from/in* Fig. 1,	
based on (data in) Fig. 1,	
can be found in Fig. 1	Fig. 1:
can be identified from/in Fig. 1	
can be observed in Fig. 1	contains
can be seen from/in Fig. 1	demonstrates
(close) inspection of Figure 1 indicates	displays
data in Fig. 1 suggest that	illustrates
(data not shown)	lists
evidence for this is in Fig. 1	plots
(Fig. 1)	presents
in/from Fig. 1 (it can be seen that)	represents
in Fig. 1 we (compare etc.)	reveals
is/are apparent from/in Fig. 1	shows
is/are clearly visible in Fig. 1	summarises
is/are evident in the figure	
is/are given in Fig. 1	
is/are shown in Fig. 1	
is/are visible in Fig. 1	
results are given in Fig. 1	
(see Fig. 1)	
we observe from/in Fig. 1 that	

*注：from表示可以从图形中的数据理解/总结/推断出它，而in表示它实际出现在图形中/在图形本身中可见。

下面是使用列表内容的例子：

- As **is apparent from Fig. 1**, the extractibility of the dehydroisoandrosterone did not change significantly.
- The data in Fig. 18 **reveal** that H3K36 can repair damaged DNA.
- The coefficients given in Table 1 **represent** mean values from both spectrophotometers.
- ZnO thin-film islands which cover the substrate surface **can be observed** in Fig. 3a.
- Figure 1 **illustrates** the responses from the three regions of the right ventricle.
- Results for individual groups are **displayed** in Supplementary Figure 5.
- As we **can see from** Figure 3, the spectrum changed significantly during the phase scan.
- The lack of correlation between IR and PLFA patterns can be seen by **comparing** Figs 1 and 5.
- An analysis of the follow-up status of the subjects is **presented** in Table 2.
- The exchange current densities are **summarised** in Table 9.

特定的/关键的结果

(ⅰ)客观的结果描述

is/was absent	change/d	match/ed
is/was constant	decline/d	occur/red
is/was different	decrease/d	peak/ed

is/was equal	delay/ed	precede/d
is/was higher, etc.	drop/ped	prevent/ed
is/was highest, etc.	exceed/ed	produce/d
is/was identical	exhibit/ed	reduce/d
is/was obtained	exist/ed	remain/ed
is/was present	expand/ed	resume/d
is/was seen	fall/fell	rise/rose
is/was unchanged	find/found	take/took place
is/was uniform	increase/d	vary/varied

注：百分比、水平、地点和数量等的数字表示(例如 **2% increase**)当然也是**客观**的结果描述。

下面是使用列表内容的例子：

- Protein expression **peaked at** day 3 after infection.
- The frequencies of the jets **were constant**, and **did not change** linearly with velocity.
- Concentration scales **decreased** with the fuel fraction.
- It was observed that the female rats **exhibited higher** levels of plasma corticosterone than male rats.
- Leaf enlargement **resumed** toward the end of the desiccation period.
- The UV-visible spectra of the purified components **were identical** to those of lutein.
- Mean SAOC values **rose** to 682 mol/L within the first hour.

（ii）对结果的主观描述：宽泛的评论

abrupt	important	reasonable	**Useful verbs:**
acceptable	in general	remarkable	emphasise
adequate	in principle	serious	highlight
appropriate	in the main	severe	reflect
brief	inadequate	sharp	resemble
broadly	likelihood	similar	
by and large	mainly	steep	**Comments:**
clear	major	striking	Importantly,
comparable	measurable	strong/ly	In fact,
comparatively	merely	subtle	Indeed,
consistent	minor	sudden	Interestingly,
distinct	more or less	sufficient	Intriguingly,
dramatic	obvious	suitable	It is noteworthy that...
effectively	perceptible	unexpected	Notably,
equivalent	poor	unlikely	Overall,
essentially	powerful	unremarkable	Significantly,
excessive	profound	unusual	Surprisingly,
extensive	pronounced	virtually	
generally	predominant/ly	weak/ly	
gradual	rapid		

注：无论 higher 在客观上是对还是错的，high是对数量/水平/金额的主观评估。

(ⅲ)对结果的主观描述：关于数量的评论

STEMM写作中用于评论数量的主观语言可以分为五组：

- **第一组**　包含增加 (increase) 大小/数量的词汇和短语：
  A considerable amount of residue remained in the pipe.
- **第二组**　包含减少 (reduce) 大小/数量的词汇和短语：
  Barely 74% of the residue remained in the pipe.
- **第三组**　包含强调大小/数量有多大/少/高/低的词汇和短语：
  The amount that remained in the pipe was even higher/even lower than predicted.
- **第四组**　包含表达大小/数量和其他结果相似或相近的词汇和短语：
  Almost half of the residue remained in the pipe.
- **第五组**　当你不想解释大小和数量时适用：
  Some of the residue remained in the pipe.

阅读下面的列表，并将每个单词和短语放在正确的组别中（那些带有星号标记“*”的单词会出现在两个组别中）。

a great deal (of)	imperceptible	quite (high/low)
a few	in some cases	quite a few

a little	just*	quite a lot
a number (of)	large	rather (high/low)
abundant	less	reasonably
appreciable	little	relatively
appreciably	low	remarkably
approximately	marginal	roughly
as many as	marked	scarcely
as few as	markedly	several
at least	minimal*	significant
barely	minor	significantly
below	moderate	slight
close (to)	modest	slightly (more/less)
considerable	more (than)	small
considerably	most	so (high/low)
easily (over/under)	much*	some
even (higher/lower)	near/ly	somewhat
exceptionally	negligible	substantial
extremely	noticeable	substantially
fairly	noticeably	to some extent
far (more/less)	numerous	under
few	only	upwards of
fewer (than)	over (half/25%)	very
greater (than)	particularly	virtually
hardly	plenty of	well (over/under)
high	practically	

答案

第一组　包含增加（increase）大小/数量的词汇和短语：

a great deal (of)	high	over (half/25%)
a number (of)	large	plenty of
abundant	marked	quite a few
appreciable	more (than)	quite a lot
as many as	most	several
at least	much	significant
considerable	noticeable	substantial
greater (than)	numerous	upwards of

第二组　包含减少（reduce）大小/数量的词汇和短语：

a few	hardly	minor
a little	just	modest
as few as	less	only
barely	little	scarcely
below	low	slight
few	marginal	small
fewer (than)	minimal	under

第三组 包含强调大小/数量有多大/少/高/低的词汇和短语：

appreciably	far (more/less)	significantly
considerably	markedly	so (high/low)
easily (over/under)	much (higher/lower)	substantially
even (higher/lower)	noticeably	very
exceptionally	particularly	well (over/under)
extremely	remarkably	

第四组 包含表达大小/数量和其他结果相似或相近的词汇和短语：

approximately	just (over/under)	practically
close (to)	little (i.e. close to none)	roughly
few (i.e. close to none)	minimal	slightly (more/less)
imperceptible	near/ly	virtually
	negligible	

第五组 当你不想解释大小和数量时适用的词汇和短语：

fairly	quite (high/low)	some
in some cases	rather (high/low)	somewhat
moderate	reasonably	to some extent
	relatively	

(iv)对结果的主观描述："高兴"字眼☺

accurate	efficient	reliable	useful
advantageous	excellent	satisfactory	valuable
beneficial	feasible	simple	viable
better	improve	successful	worthwhile
effective	precise	superior	

(v)对结果的主观描述：关于频率的评论

频率修饰语可帮助读者评估结果。例如，如果研究人员说每次(every time)执行特定测试都会产生特定结果，那么显然这是一个非常可靠的结果。如果它通常(generally)会发生，这仍然是可靠的，但可能可靠性就没那么高了。如果作者仅声明某事发生(X occurred)而没有使用频率修饰语，那么读者可能无法正确评估结果，或者根本无法评估结果。

STEMM写作中用于评论频率的评估语言可以分为十个等级。想象一下，你想在星期一的早晨找到一位同事，并且想知道是否可以在图书馆里找到她。

1. 如果她总是(always)在星期一的早上去图书馆，那么你今天可以在那里找到她。

2. 如果她通常(generally)在星期一的早上去图书馆，你会期待今天能在那里找到她，但是如果她不在那里，你会觉得惊奇。

3. 如果她经常(often)在星期一的早上去图书馆，那么你很

可能今天可以在那里找到她。

4. 如果她大概率(more often than not)会在星期一的早上去图书馆,你可以到那里去找她,但是她可能今天不在那里。

5. 如果她星期一的早上有时去有时不去(as often as not)图书馆,你也许今天能找到她——也许不能。但是无法预测是否能找到她,因为两者的概率相等,她去与不去的概率相等。

6. 如果她有时(sometimes)会在星期一的早上去图书馆,她可能今天会在那里,但是如果不在也不奇怪。

7. 如果她偶尔(occasionally)在星期一的早上去图书馆,那么她也许今天会在那里,但是不太可能。

8. 如果她极少(rarely)在星期一的早上去图书馆,那么她很可能今天不在那里,所以不要去那里找她。

9. 如果她几乎不(hardly ever)在星期一的早上去图书馆,那么不要期望她今天在那里,如果你在那里看到她会感到很奇怪。

10. 如果她从来都不(never)在星期一的早上去图书馆,那么她今天不在那里。

下面的列表按频率从高到低的顺序显示了十个等级。但是,要注意的是关于频率的陈述往往是主观的:说某件事频繁发生(occurred frequently)可能是相对于它预期发生(expected to occur)的频率。例如,如果早期的研究表明特定的结果不太可能发生,但是在你的研究中,该结果在22% 的情况下发生,那么你可能称该结果为频繁发生(frequent occurrence)。另一方面,如果早期的研究表明很可能会发生某种情况,但在你的

研究中仅在22% 的情况下发生，那么你可能会称这种情况很少（rare occurrence）发生。

1	each/every time without exception on each/every occasion always invariably
2	habitually as a rule generally normally usually
3	regularly repeatedly frequently often commonly
4	more often than not
5	as often as not (**neutral frequency**)
6	sometimes on some occasions at times
7	occasionally now and then from time to time
8	infrequently rarely seldom

9	hardly ever barely ever almost never scarcely ever
10	on no occasion not once at no time never

阅读你的目标文章，你会发现我们很难找到**客观**描述结果的语言，更易找到那些对结果的**主观**描述。当有客观语言出现时往往有主观修饰词的加入。例如，我们更多地发现像 slightly lower（略微低一些）或 much lower（低很多）这样的表达而不是单独的 lower（低的）这样的客观描述。正如前面提到的，这是因为对结果的客观描述对读者来说跟他们自己看图像没什么很大的差别。

如果你很难区分**主、客观**语言表达，请记住描述一个水平或数量比另一个更高（higher）是客观的事实；而描述其本身是高的（high）则为主观的评价。

这是一些关于结果的主观描述的例子：

- **Minor** clinical abnormalities of nerve function were **fairly common** in the rheumatoid group.
- **Significantly**, we found that the activation energy for oxidation

in both gases is **much** higher.

- The yeast samples were, however, found to be **relatively** poor sources for dehydrogenases.
- Overall, there was a **close similarity** between the biological activity and immunoreactivity of plasma ACTH in the three specimens.
- The impacts of other micronutrients were **in general quite** weak.
- Extracts prepared at 0.3 M NaCl were **only slightly** active while extracts prepared at concentrations of 0.35 0.5 M NaCl were **quite** active.
- **In several cases** the chromaticity of exposed skin was **noticeably** different from that of unexposed skin.
- **Notably**, the non-magmatic IAB and IIICD irons display overlapping Zn contents, in accord with the previous conclusion that these two groups are **closely related**.
- The solenoid valve **frequently** generated a more accurate pressure response.
- Ageing of the alloy showed that the YS was **virtually** unchanged.
- Survival rates in the steroid treated patients were **essentially identical** to those of the placebo treated patients.
- The magnitudes of the ionic displacements were found to be **extremely** large.
- These abnormalities were **invariably** preceded by symptoms of toxicity.
- The calculated Young's modulus was **almost** constant in the

area above the nucleus.

- There was a **striking** difference in the image contrast between the gradient echo image and the spin echo image.
- The colours of the two lights were found to be **extremely similar**, if not identical.
- The amount of dye adsorbed on gold was **even higher** than that on the silver particles.
- Breaks in nanotubes were **rarely** observed following specimen cutting.

将你的结果与其他结果/建模结果/预测结果比较

as expected,	**Verbs**
better than	accord with
broadly similar to	align with
comparable to	compare well with
comparatively (low/high etc.)	confirm
consistent with	contradict
contrary to	correlate with
(effectively) the same as	corroborate
(essentially) identical	deviate from
in accordance with	differ
in (good) agreement with	disprove
in contradiction to	mirror
in contrast to	prove
in line with	refute
much the same as	reinforce
not dissimilar	resemble

not unlike	substantiate
relative to	support
similar	validate
unlike	verify
well known	

注：动词的时态可能会随着时间的推移、知识和信息的发展而变化。这意味着从五年前开始发现某篇文章中出现了类似（X has been found to occur[4]）的句子，现在可以将其写成X occurs[4]，甚至只是 X occurs。

添加例如看起来（appear to）、似乎（seem to）、倾向于（tend to）这样降低风险的短语，可以传达作者想要表达的确定性的程度（请参见第3.4.2节的“结果的含义和解释”、第3.5节的“确定性连续统”和第4.5节的“情态动词”）。

下面是这些短语应用的具体实例：

- These data **appear to confirm** the prevalence of Vitamin D deficiency in the normal population.
- Thus, the new data from the Human Genome Project **tend to corroborate** the result in Fig. 2.
- Our results **differ from** those in the literature, and **agree better with** published CDF data.
- These new results are **consistent with**, and **substantially more** sensitive than, previously published anisotropy measurements.
- The bubble growth predicted from numerical analysis **compares well with** that obtained in this work.

- The location of the gas concentration in the simulation is **in remarkably good agreement with** that of the observed dust lanes.
- The accumulation rates **we observed are in line with** those reported from sulfate incorporation methods.

问题和议题

minimise problem (*no big deal*)	**minimise responsibility (*not my fault*)**	**maximise good aspects (*I got good stuff anyway*)**
a preliminary attempt	as far as possible	despite this
acceptable	difficult to (simulate)	fairly well
did not align precisely	hard to (control)	nevertheless
immaterial	impossible	quite good
insignificant	impractical	reasonably robust
less than ideal	inevitable/ly	
less than perfect	(it was) difficult to	**....and focus on a**
marginal/ly	(it was) hard to	**solution or a reason**
minor deficit	limited by	
negligible effect	necessarily	future work should...
non-ideal/not ideal	not examined	future work will…
not complete	not explored in this study	further work is required
not identical	not investigated	(is) currently in progress
not perfect	not possible	(is) currently underway
not significant	not within the scope of this study	[this] was because…
of no consequence	unavoidable	

of no significance	unexpected	
reasonable	unfortunately	
slightly (disappointing)	unpredictable	
(some) imperfection/s	unreachable	
somewhat (problematic)	unworkable	
technicality		
unimportant		

下面是应用的具体实例：

- **Inevitably**, considerable computation was involved.
- **Unfortunately, it was not possible to** quantify the extent of Zn loss from the experimental charges.
- Only a brief observation was feasible, **however**, given the number in the sample.
- **Although** centrifugation could not remove all the excess solid drug, the amount remaining **was negligible**.
- Solutions using (q=1) **differed slightly** from the analytical solutions.
- **A further study will** examine dc-dc converters to determine whether these efficiencies can be achieved in practice.
- While the anode layer was slightly thicker than 13 μm, **this was a minor deficit**.
- **Future work should** therefore include numerical diffusion effects in the calculation of permeability.
- This type of control saturation is fairly common and therefore **of no significance**.

结果的含义和解释

（阅读下面第3.5节中的示例句子：确定性连续统）

could* be explained by	apparently,
could* be interpreted as	evidently,
could* be seen as	in part due to…
	it appears that…
likely	it could* be inferred that…
perhaps	it is [fairly/abundantly] evident that…
possible	it is [very/highly] probable/likely that…
presumably	it is logical that…
probably	it is thought/believed that…
unlikely	it may (well) be that…
	it may be concluded that…
Verbs	it may/can be assumed that…
deduce	it seems (very/highly) probable/likely that…
imply	it seems that…
indicate	it would seem/appear that…
infer	the evidence suggests that…
mean	there is evidence to indicate that…
signify	this implies/seems to imply/may imply that…
suggest	this is (compelling) evidence for…
	this is indicative of…
	this seems to suggest that…
	we have confidence that…
	we propose that…

*注：could 可以被 may 或者 might ——有时也可以用 can ——代替。下一单元将介绍如何使用这些情态动词。

3.5　语言和写作技巧：确定性连续统

结果部分显示了你所做的事情与发现的事情/发生的事情之间的关系。有时你有足够的证据说明你所做的事情肯定会导致一个特定结果的发生；在其他时候，你可能确定你所做的事情会导致一个特定结果的发生，但是你没有足够的证据。你该如何向读者传达这种程度的信心呢？

结果部分的另一个功能是指明你的发现可能产生的影响。有时候你可能会确定你的发现暗示某些东西，但是你的结果无法证明这一点。确定性连续统包括从推测一直延续到确凿证明的不同程度。你对结果的信心适合该连续统的哪个位置？你有多么自信？你有多么肯定？你对这些信心有多坚定，以及如何与读者交流？

3.5.1　选择一个能准确反映你所描述的因果关系的动词

有些动词传达直接/强有力的因果关系（cause、produce、be due to）。

有些动词提及部分原因（contribute to、be a factor in）。

有些动词暗示了一个有因果关系的过程（lead to）。

有些动词提及因果链中的起因（originate in）。

有些动词提及一种产出，而不是一种效果（produce/yield）。

有些动词沟通一种间接/微弱/内含的因果联系（be related to、be linked to）。

(be) a factor in	accompany	govern
(be) a/the cause of	account for	influence
(be) a/the consequence of	affect	initiate
(be) a/the result of	arise from	lead to
(be) ascribed to	cause	originate in
(be) associated with	contribute to	produce
(be) attributed to	create	result from
(be) connected to	drive	result in
(be) due to	generate	yield
(be) linked to	give rise to	
(be) related to		

在某些情况下，原因和结果的位置由动词确定。例如，在X produced Y 中，X是原因而Y是结果；在X originated in Y中，X是结果而Y是原因。在其他情况下，例如X is linked to Y 中，作者可能只想表明X和Y以某种方式连接。这些动词（linked to、connected to、related to）没有指定哪个是原因，哪个是结果——甚至也不表示两者之间的联系必然是因果关系。

冠词和介词也会影响因果关系陈述的含义。例如：

- 如果陈述内容为X is **a** cause of something或者X is **a** result of something，这表示其他因素也有关，而如果 X is **the**

cause of something或者X is **the** result of something，这表示X是唯一的原因或者结果。

- X results **from** Y意味着“X is a /the consequence of Y.”。X results in Y意味着“Y is a/the consequence of X.”。

3.5.2　选择合适的动词时态

在第一单元（**1.5.1动词时态的选择**）已经指出时态选择可以传达作者对结果的永久价值或永久事实的信心。例如，在下面的句子中：

We found that sunbathing **was related to** cancer.
We found that sunbathing **is related to** cancer.

过去时（was related to）仅描述作者在研究中发现的内容，这些发现与该研究有关，并没有以普遍接受或公认的事实来呈现。相比之下，选择现在时（is related to）反映了有足够可靠的信心构成永久事实。

3.5.3　添加可降低风险的语言

添加如 it appears that.../there is evidence to indicate that… 这样的短语来降低声明的风险。其他可降低风险的语言包括：

- 频率修饰词，例如：often、commonly（参看第3.4.2节）。
- 数量修饰词，例如：in some cases、in virtually all cases（参看第3.4.2节 ）。
- 情态动词，例如：may、might（参看第4.5节）。

总而言之，第3.5节就整个确定性连续统提出了语言建议。一旦弄清楚了关于结果的信心在连续统中的位置，以及你对该信心的承诺程度，那就可以选择适当的语言。你可以使用第3.5节中的语言来表达所有程度的自信和确定性，从**绝对确定**到**对结果可能产生的影响有非常不确定的建议**：

- We **find/found that** sunbathing **causes** cancer.
- We **find/found** that sunbathing **may** cause cancer.
- We **find/found evidence to suggest that** sunbathing **may be related to** cancer.
- **It appears therefore that in some cases**, sunbathing **may have been related to** the onset of cancer.
- **The evidence points to the possibility that in some cases, excessive** sunbathing **may have contributed to** the onset of **certain types of** cancer.

下面以最近发表的研究论文中结果部分的例子来展示如何使用语言来传达不同程度的自信和确定性：

- The temperature of the Ti target **appeared to be somewhat** critical.

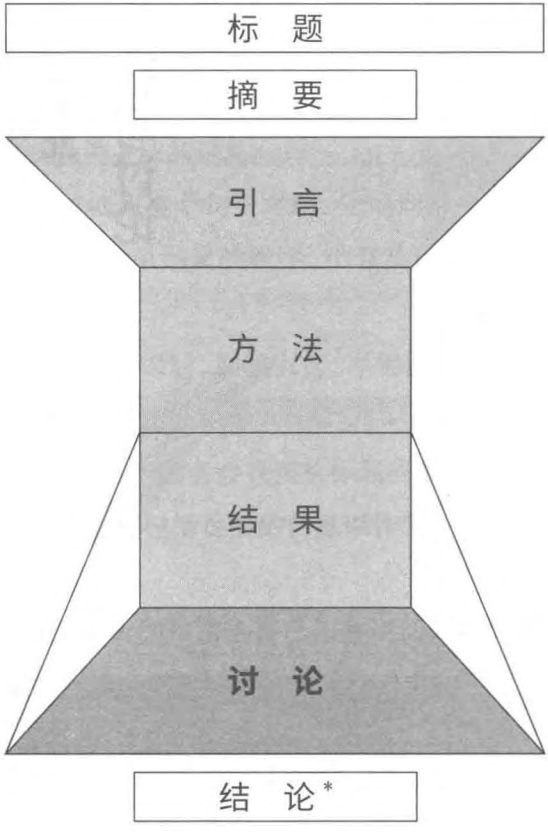
- **It seems therefore** that the plating solution **probably affects** the ceria.
- The identified pattern **is potentially indicative of** altered cellular processes.
- **It is** therefore **reasonable to suppose that** the duplexity of pitch **is a reflection of** duplexity in the auditory process.
- Therefore, from these results **it could be inferred that** the sealant did not improve the coating protective performance.
- The Zn correlations of the samples (Figs. 3, 4a–c) are hence **most plausibly explained by** indigenous processes.
- This **provides strong evidence for** the importance of NADW in glacialinterglacial climate change.
- We **speculate that this reflects** the increase in blood pressure that is known to occur in some of these patients.
- **It is therefore possible that** the correlations reflect variable contamination.
- **This suggests that** the polysulfide **may** undergo more complicated electrochemical reactions with **likely** involvement of more polysulfide species.
- **These results indicate that** EGFR mutations **may** predict sensitivity to gefitinib.
- The differences in recurrence rates increased over time, **suggesting that** there is a carryover effect.
- **It seems, therefore, that** spontaneous reactivation of viral replication **may be** a rare event in HBsAg-positive patients.
- Therefore, **it appears that** the plating solution **probably affects** the ceria within the anode composite.

下面是编写成功的结果部分的窍门：

- 检查目标文章结果部分的长度、小节和小标题的格式。
- 检查方法部分通常出现在目标刊物中的位置，以及这对结果部分的内容产生的影响，即结果部分中要包含多少有关方法的信息。
- 在开始创建整个句子之前，先规划整个结果部分的结构：决定显示结果的顺序、小节及其标题，以及图像的位置。
- 获得研究结果后，回顾引言，以确保引言中所述的论文目的与研究结果相符。
- 请记住，很多读者会直接从标题或摘要转到结果，因此结果部分要包括足够的信息以作为一个独立的部分发挥功能。
- 请记住，数据和结果自己不会说话。在叙述中有逻辑地和自然地呈现结果以引导读者走向你想要得出的并希望读者分享的诠释和结论。
- 检查目标文章中结果部分的评论和评价性语言的数量。
- 决定你的结果以及结果的含义在确定性连续统中适合的位置，并选择代表该位置的语言。

第四单元
怎么写讨论





* 有些期刊在这一部分使用结论一词的单数形式，有些使用复数形式，但这似乎并不能反映所得结论的数量。

4.1　怎么写讨论

如第3.1节所述，结果部分与讨论部分分开处理，并且讨论部分与结论分开处理。这是为了确保你可以训练自己去识别并由此创建每个部分的特征组件，无论是将它们分开还是合并。

将讨论包装在叙述中

一个成功的讨论部分的关键是有一个向前移动的、组织良好的叙述包装（见引言：**叙述性包装是什么**？），它耐心地、有逻辑地和明确地引导读者从结果到结论。

在计划讨论之前，先从研究中退出来并考虑一下其**主要价值或贡献**，这是有帮助的，例如：

- 该研究是否获得了与其他研究完全相同或相似的结果，但使用了比现有方法更好的改良的或全新的方法？**在这种情况下，主要贡献可能是方法**。
- 该研究是否获得了比其他研究更好（例如，更准确）的结果？**在这种情况下，主要贡献可能是结果本身**。
- 这项研究是否改变了游戏规则，即它是否开创了新的研究方向/推翻已有工作？**在这种情况下，该研究的主要贡献可能是它对文献/研究界的影响**。

- 该研究是否确定或创建或扩展了应用？**在这种情况下，该研究的主要贡献可能是它对工业界或实际应用的影响。**

上述例子设定了一个明确的“目标”，该目标可以通过叙述明确地传达给读者，并确保给读者留下深刻印象的关键信息不会在不相关的细节中丢失。研究人员阅读一篇研究文章的时间通常不超过30分钟，因此尽可能明确地优先考虑和传达主要价值或贡献是至关重要的。

4.2 建立你自己的模板

虽然讨论中各个组件的顺序是灵活的，但组件本身相对简单明了，并遵循研究文章的对称“形状”。引言从相当宽泛的语言开始，逐渐缩小到方法/结果，而讨论则从方法/结果转向更广泛的焦点。回顾下面通用的引言部分模板是一个很好的起点。

通用引言模板（来自第一单元）	
1	论证研究主题/领域的重要性 给出背景事实和信息 给出研究领域的一般性问题/当前研究焦点
2	呈现已有研究和/或当前的科研和贡献：研究“地图”

3	确定已有研究工作的空白 描述你要解决的问题 呈现你的动机和/或假设 确定研究机会
4	描述当前的论文，有时会提及目标/结果/方法/结论，并经常包括“高兴”的字眼

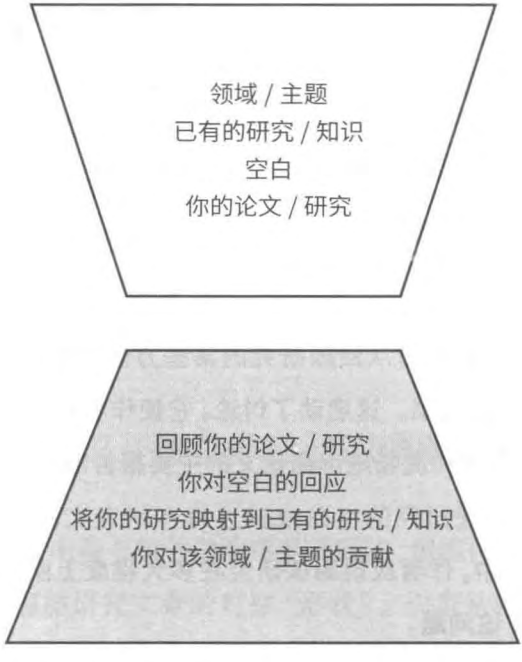
在引言的第4块中，作者描述了当前的论文或研究，在引言和论文的中心部分之间提供了一个有逻辑性的和流畅的接口。**在讨论中，作者通常从回顾研究的某些方面开始，例如研究目标/空白或关键结果。这启动了讨论，它使作者能够反向创建接口，并有逻辑地和流畅地远离论文的主要报告部分，转向结论。**

第3块呈现研究的空白或描述将在研究文章中解决的问题。**在讨论中，作者应说明该研究在多大程度上回应了这一差距或解决了该问题。**

第2块展示了一个研究地图，其中确定了关键的研究和贡献。**在讨论中，要把当前的研究定位于该研究地图的相关位置。**

第1块提供了外部世界和研究文章之间的接口，通过确定主题的重要性、提供背景信息和/或建立当前研究焦点，使读者准备“跨越边界”进入文章。**讨论的结尾创建了一个类似的接口，为读者退出文章并将关键信息传递给外部世界和/或研究世界做出准备。**

下图显示了序言和讨论是如何相互呼应的：



4.2.1 练习1: 用目标文章去建立一个讨论模板

本单元鼓励你使用逆向工程的方法构建一个模板，**这个模板展示你所在领域当前与你所涉及的研究主题相关的研究文章中的讨论部分**。

记住上述的对称图，通过对目标文章的讨论部分进行逆向工程的方式来生成一个潜在组件的列表。要创建此列表，请在

至少**两篇**目标文章的讨论部分对每个句子功能进行简要描述。完成以后，阅读下面简化的讨论部分的分析，然后将你的列表与第4.2.3节的通用模板进行比较。列表中的一些组件类似于通用模板中的组件；其他组件针对你正在处理的研究主题、你所在领域的当前惯例或你正在撰写的文章类型。将你的列表与通用列表合并，为当前研究领域的讨论部分生成一组健全的模板组件，并根据需要进行调整或更新。

An assessment of post V2D sequelae in disease survivors

Discussion

1 *To our knowledge, the study reported here is the largest study of patients who have survived the viral disease V2D, and provides a key contribution to the understanding of post-V2D health issues following infection.* ***2*** *Our study is the first to record a wide range of post-V2D complications, and provides clear evidence that survivors who experience a given symptom during V2D infection often present with a closely-related symptom within 300 days following infection.* ***3*** *This was particularly true in relation to some specific complications, such as hearing loss (Fig. 1).* ***4*** *A key finding was that some post-V2D health issues appear to be age-related: infants below the age of 2 experienced post-V2D cardiac complications far less frequently than adults irrespective of the severity of such complications during the*

*acute stage (Figs. 2a and 2b). **5** This has clear implications for post-V2D-related public health management, and suggests that when resources are limited, cardiac screening priority could be given to adults.*

***6** Some questions remain; for example, anecdotal evidence collected from patients during the study period suggested that there was a positive correlation between viral load at the acute stage and the severity of post-V2D health issues. 7 However, this was not addressed formally during the study. **8** Although many studies have quantified viral load during acute V2D[17,22,27,33], a direct comparison with the post-viral period would provide valuable data to support such a correlation.*

***9** V2D RNA is known to persist in the bodies of survivors long after disease onset[30,42], and the results of our study provide compelling evidence that follow-up of all convalescent patients is warranted for at least 10 months after discharge from hospital. **10** In addition, the study provides evidence-based guidance for pragmatic public health recommendations that include age-related follow-up and treatment of post-V2D complications.*

4.2.2 答案

句1："To our knowledge, the study reported here is the largest study of patients who have survived the viral disease V2D, and

provides a key contribution to the understanding of post-V2D health issues following infection." **作者明确指出该研究对已有文献和知识的贡献。**

▶ 据我们所知（to our knowledge）听起来是不是有点弱？我不是应该知道它是不是最大的研究吗？

在句1中，作者包含了据我们所知（to our knowledge）的短语，以防意外忽略了某些研究。即使在尽一切努力搜索所有相关期刊之后，也很难绝对确定到目前为止没有人进行过一个特定类型的研究。在即将提交论文之前，你应该尽可能彻底地使用大量的关键字来检查当前的研究。你从网上得到的信息只会和你在查找信息的过程中的技巧一样好，而且在这样的句子中犯错是不专业的。研究的发展和发表速度非常快，在提交到你的目标期刊/接受同行评议/重新起草并重新提交所需的时间内，类似的研究可能已经发表了。如果是这种情况，请不要惊慌。你可能只需要回顾和澄清你的研究与其他研究之间的异同点，以便区分和确定你对该主题的贡献。

▶ 为什么要这样开始讨论？

在方法和结果小节中，作者可以从关于该小节内容的一般陈述开始，创建一个对读者友好的入口。例如，在第三单元中，我们看到作者有时以关于结果的一般陈述开始结果部分。然而，尽管一些讨论部分确实以关于讨论部分将包含什么的一般

性陈述开始，但这并不常见。在大多数情况下，讨论用以下两种方式之一开始，这两种方式都为文章的目的地或关键信息创建了一条清晰的路径。

1. 确认该研究的成就、贡献或潜在应用。

研究的成就、贡献或潜在应用和影响是关系到研究基金和专业声誉的关键因素，这推动了在摘要和讨论/结论中对影响进行清晰和明确陈述的趋势。在讨论部分开始时确认成就、贡献或潜在应用，建立一个主题框架，将其影响牢牢地留在你自己和读者的脑海中。研究成就对你和你的同事来说可能是清楚明白的，但它对所有读者，尤其是跨学科的读者来说都并非如此。

在讨论部分的开始就确认研究的影响或价值的另一个原因是：读者可能以一个简单的问题来支配对文章的快速和高度的选择性阅读，例如**这项研究到底实现了什么？或者这项研究如何影响我自己的研究？**该读者可能是一个跨学科读者，对研究的技术方面知之甚少，他们的问题是更实际的：**这些研究的潜在应用是什么？**通过陈述成就、贡献或潜在应用来开始讨论部分是对此类信息冲浪的强有力的回应。

2. 通过下面这几点来“重新启动”读者：

- 回顾相关的背景事实信息。
- 回顾文献中的差距或研究目的。

- 回顾结果或方法的关键特征。

正如前言中所述，大多数读者不会从一开始就线性地阅读整篇研究文章。很多人只阅读标题、摘要、讨论/结论，这些部分需要能够成为切实可行的独立沟通单元。此外，即使读者到此为止已经阅读了整篇文章，也不能假定他们已经有和作者相同的注意力和专注度阅读了这篇文章，或者读者能记住他们所阅读的所有内容。基于这些原因，作者通常会在讨论部分开始时通过重新启动与研究的相关方面来重新定位或重新引导读者。

▶ 如果我决定“重新启动”读者，我该如何选择要回顾文章的哪一部分呢？

成功的讨论依赖于一个可控的叙述过程，它清楚地确定了研究的主要价值，并合乎逻辑地、有效地将读者带到结论。这个叙述从讨论的第一句话就开始了，因此思考你的研究的主要成就或贡献（参见第4.1节）可能会帮助你决定首先关注什么。例如，如果你的研究的主要贡献是你对现有方法所做的更改或修改，那么你可以首先回顾与该现有方法相关的困难，或者回顾新方法的主要特征和相对优势。如果你的研究的主要成就是你的结果比已有结果更准确，或者展示了新信息，那么你可以从回顾你的结果开始。如果你的研究的主要贡献是对引言所指明的研究的空白或问题提供了强有力的回应，那么你可以先从

回顾那个空白开始。

在这部分使用与你选择回顾的内容相似的句子，这将为读者提供回声，帮助他们回想起或找到那部分。和往常一样，通过查看目标期刊中的作者如何开始讨论部分来进行逆向工程。

句2："Our study is the first to record a wide range of post-V2D complications, and provides clear evidence that survivors who experience a given symptom during V2D infection often present with a closely-related symptom within 300 days following infection." **作者根据已有文献确认该研究的具体新颖特征，并概述结果。**

▶ 我的研究如何纳入已有文献?

将研究工作映射到现有文献和知识领域以确认你的"产品"在研究"市场"中的位置。映射是讨论的核心功能，它展示了这项研究工作如何推动研究课题的发展。例如，你的方法可以比已有方法更快地生成结果，这可能会影响以前那些方法的地位及其应用。你的研究可能会证实早期研究中获得的结果，这将验证那项早期研究工作，也许还有其结论。或者，你的研究可能与早期研究中得到的结果相矛盾，这会引发对早期结果的质疑，并可能会为研究确立新的方向。

讨论中应该有多少引文?

引文对于定位你的研究并将你的贡献与早期的知识分开至关重要。通过简单地平均目标文章讨论部分中的引文次数，你可以粗略了解特定期刊中既定主题的正常引文情况。留意每个引文的功能、位置、特定引文的重复频率，以及每个引文是如何支撑讨论的。每个引文都应该是相关的，并且应该通过某种叙述向读者明确说明其相关性。

在下面的示例中，那些发展了当前研究与现有研究/知识之间关系的句子已从讨论中“提升”，以展示如何围绕引文构建叙述手法。查看句子开始的方式特别有用，因为这些短语发展了当前研究与现有文献和知识之间的关系。

We first demonstrated that _____ , consistent with the literature (2, 4, 45).

In particular, _____ was shown to _____ (40).

Regarding this concern, Söderholm *et al.* (45) reported that _____.

In contrast, O’ Malley *et al.* (34) reported _____.

To verify this assumption, _____was used, as previously depicted for _____ (7, 38).

Using _____ , Lai *et al.* (27) demonstrated that _____.

As a result, _____ , as determined by Rahli *et al.* (36).

In line with this, a recent in vivo study by Wrzosek *et al.* (47) clearly showed that _____ .

Likewise, the role of _____ has been proposed by Gaudier *et al.* (18).

***Edited extract from*: Stress disrupts intestinal mucus barrier in rats via mucin *O*-glycosylation shift: prevention by a probiotic treatment[q]**

Discussion

IBS is a multifactorial disease, frequently associated with psychological distress and characterized by altered gastrointestinal motor function, viscerosensitivity, and, more recently, low-grade inflammation and impaired intestinal barrier function. Noteworthy, a positive correlation between increased intestinal permeability and visceral pain has been shown in IBS patients (2). Since stressful life events contribute to the IBS symptom exacerbation, the most commonly animal models reproducing "IBS-like" symptoms result from the use of central nervous system-directed stressors (33). In this study, WAS was used as a reliable model of chronic psychological stress. **We first demonstrated that a 4-day WAS induced intestinal epithelial barrier impairment, consistent with the literature (2, 4, 45). In particular, for WAS, colonic permeability increase was shown to occur from the third day of a 5-day stress application (40).** [...] information regarding the effects of chronic stress on mucus barrier, a major actor ensuring intestinal barrier integrity, remains

scarce in IBS-like models. **Regarding this concern, Söderholm *et al.* (45) reported that a 10-day WAS decreased the number of mucus-containing goblet cells in the ileum and colon and increased bacterial adhesion.** A reduction in the goblet cell number was also depicted in the duodenum of maternally deprived rats (15). **In contrast, O'Malley *et al.* (34) reported an increased colonic mucus secretion and number of goblet cells in maternally deprived rats.**

[...] Larsson *et al.* (29) depicted in active phase UC patients an altered Muc2 *O*-glycosylation profile, which was correlated with inflammation severity. Such profile was characterized by changes in the rate of glycanic structures and decreased overall sulfation (29). More recently, the same group showed that mucus of patients with active UC was penetrable ex vivo to 2-μm-sized fluorescent beads (24), suggesting that biochemical changes in mucins impact mucus "penetrability," resulting in a loss of barrier integrity. In this framework, we hypothesized that structural *O*-glycan alterations induced by stress may in turn change the physical properties of intestinal mucus. **To verify this assumption, AFM was used, as previously depicted for the characterization of conformational and hydrodynamic properties of individual mucin polymers (7, 38).** [...] At the nanoscale, mucus has been described as a heterogeneous mesh network of mucin fibers (41, 44). **Using engineered nonmucoadhesive nanoparticles, Lai *et al.* (27) demonstrated that the mesh structure of human cervicovaginal mucus was more open (average pore size**

340 ± 70 nm, range ~50–1800 nm) than the 15- to 100-nm pore size expected. […] **As a result, the mucus network may exhibit higher pore size and increased permeability since, for a given porosity (in the range 0–0.7), the permeability of a fibrous network is directly proportional to the square of the fiber diameter, as determined by Rahli *et al.* (36) for porous media made up of randomly packed monodisperse fibers.**

[…] Direct or indirect regulation of glycosylation enzymes may be a possible candidate. **In line with this, a recent in vivo study by Wrzosek *et al.* (47) clearly showed that commensal bacteria, such as *Bacteroidetes thetaiotaomicron*, could directly influence goblet cell development and enhance expression of genes encoding host enzymes involved in mucin glycosylation, like sialyltransferase *st3gal4*.** In an in vitro study, Freitas *et al.* (17) deciphered the ability of a heat soluble factor from *B. thetaiotaomicron* to modulate specifically the galactosylation pattern of HT29-MTX cells, through a mechanism involving posttranslational induction of glycosyltransferase activities. **Likewise, the role of other bacterial metabolites, and particularly butyrate, as factors directly acting on mucin glycosylation has been proposed by Gaudier *et al.* (18).** […]

▶ 我可以在讨论中引用我前面没有提及的研究吗?

这取决于引文与你讨论的相关性。如果引文是另一项研究中的数据诠释,或者另一项研究中所讨论的应用与本论文的讨

论部分高度相关，则可以在这里首次提及。

我该用哪种时态？

对最近的目标文章使用逆向工程分析将会得到有关讨论部分的动词时态的有用数据。例如，使用一般现在时提供背景事实信息的句子数量将会让你很好地了解这些信息对于你正在进行的同类研究和目标读者群是多么常见和必要的。

为什么第 2 句是使用一般现在时？

动词时态的使用通常是一个可供选择的问题，它反映了作者想要传达的内容，而不是事件发生的时间。第一单元（第 1.5.1节动词时态的选择）中指出，时态选择可以传达作者对陈述内容的永久价值或永久事实的信心。例如，如果我们用过去时态重写上面的句子2（"...clear evidence that survivors who experienced a given symptom during V2D infection often **presented** with a closely-related symptom following infection."），这句话将仅仅是描述作者在他们的研究中发现的内容。使用过去时态意味着结果与该研究相关，而不是作为永久的、独立的事实呈现。相比之下，选择一般现在时（"...clear evidence that survivors who experience a given symptom during V2D infection often **present** with a closely-related symptom following infection."）使陈述更有力，反映出作者认为调查结果足够可靠，可以构成一个永久的、独立的事实的信心。

与大多数情况下关于科学学术写作惯例的决定不同，选择使用哪种时态来传达结果和结果的含义并不能真正通过查看目标文章作为指导来解决。在大多数情况下，作者需要自己决定数据是否足够强大以支持一般现在时态的陈述。请注意，即使选择了一般现在时，也可能包含一些免责声明以减轻作者对陈述的信心或承诺，例如在本句中：often present with a closely-related symptom.

句3和句4中："This was particularly true in relation to some specific complications, such as hearing loss (Fig. 1). A key finding was that some post -V2D health issues appear to be age-related: infants below the age of 2 experienced post-V2D cardiac complications far less frequently than adults irrespective of the severity of such complications during the acute stage (Figs. 2a and 2b)." **作者回顾结果并对其进行了评论**（particularly true… a key finding was… appear to be…far less）。

▶ 据说讨论部分不应重复结果。我应该提及结果吗？如果是这样，如何在不重复的情况下提及它们？

结果和讨论部分之间有相当多的重复。例如，正如我们在第三单元中看到的，在结果部分，大多数作者都对结果的**建议**或**暗示**进行了评论，尽管理论上这属于讨论部分。同样，就算有可能，在不提及或重复关键结果的情况下，在讨论部分发展

其具体含义也是困难的，因为这些结果提供了支持和验证这些含义的证据。然而，在讨论部分重复甚至重述关键结果是不够的，而应该从结果部分进行延伸。研究的目的不仅仅是得到和描述结果，而是在现有知识背景下理解这些结果，并对它们的含义的合理性和实用性进一步说明，例如，这些结果对上下文意味着什么，以及它们如何与引言中陈述的原始问题、假设或目标相联系。**阐述出你的结果是结果部分的核心功能，继续讨论或探索它们的含义是讨论部分的核心功能。**

使用逆向工程分析成功发表的论文将帮助你避免落入“我应该还是不应该？”的困境，使你的注意力集中于成功的作者到底做了什么。如果你检查目标期刊的讨论部分，你会发现几乎所有的讨论部分都会重复或总结关键结果，以确定其解释和内涵。

如果我对结果没有信心怎么办？

科学研究从不会到达对某个特定主题完全知晓的终点。大多数科学作者都会小心谨慎，不做出过分自信的概括，因为下一步的研究将会完善和发展前面的研究工作，依此类推。然而，同样重要的是不要为了安全起见而低估你工作的影响，因为这会破坏和削弱它。

关键是你需要根据这些结果和现有的知识来确定你对结果意义的确定性程度。选择合适的语言在这里至关重要。如果你用来讨论结果含义的语言不匹配或不能反映结果的力量，这很可能是同行评议阶段的批评点。你可以选择表明你对结果有信

心的语言，或者选择告诉读者你正在推测其含义的语言（参见**第3.5节**）。

▶ 为什么句3和句4用过去时？这与确定性程度有关吗？

如上所述，作者选择反映适当的可信度或确定性程度的动词时态。

这些句子中的动词时态如何影响读者的反应？

A key finding was that some post-V2D health issues appeared to be age-related…

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在目标文章的讨论部分突出显示动词，并考虑时态如何影响你作为读者的反应，这是一个好主意。动词时态看上去与结果的影响力“匹配”吗？

句5：“This has clear implications for post-V2D-related public health management, and suggests that when resources are limited, cardiac screening priority could be given to adults.” **作者提及从结果中产生的潜在应用。**

▶ 在讨论结束时提及潜在的应用不是更好吗？

论文的结尾通过对将来研究的建议与研究世界衔接，并通